



BEFORE

Abundant raw materials and significant technological advances created the conditions for industrialization in Britain, which then spread to Europe and the US.

BRITAIN FIRST

Access to cheap imports, such as cotton, silk, and wool, from its colonies gave Britain an advantage in being able to industrialize faster than the rest of Europe.

RAW MATERIALS

IRON MASTERS

Two key innovations made by British ironworks owners contributed to industrialization. First, in 1709, Abraham Darby's use of **coke** (processed coal that burns at **much higher temperatures**) instead of charcoal reduced the cost of ironmaking, due to coke's relative abundance. Secondly, Henry Cort's "puddling" and "rolling" processes of the 1780s produced a **purer form of iron** that was **malleable** and more useful to industrial processes than the brittle "pig iron," or "cast iron."

TRANSPORTATION NETWORK

Transportation was another vital element during industrialization. Raw materials had to be moved to the point of manufacture, and finished goods went to cities and ports for distribution. Britain had a network of navigable rivers, which it enhanced by building canals. The Bridgewater Canal, Manchester, cut in 1761, was followed by other canals that were connected to major rivers.



BRIDGEWATER CANAL

4,250 MILES (6,800 km) of inland waterways were built in England between 1760 and 1800.

FOLLOWING THE SEAM

The **rich belt of coal** that runs through the industrial north of England also runs through Silesia, the Ruhr Valley, and across the Atlantic to Pennsylvania, which would also become **heavily industrialized** when that coal was exploited.

FREEDOM TO TRADE

An empire unhindered by continental Europe's **internal frontiers** and **trade barriers** gave Britain a trading advantage. Slave plantations in America and the West Indies **<< 280–81** had given Britain a supply of cheap cotton, and when slavery was abolished in 1807, trade with the colonies continued. Even after the American Revolution **298–99 >>** the United States was still an important market.



The Industrial Revolution

The world we live in today owes much to the industrialization that took place first in Britain, in the late 18th century, and swept across Europe and North America during the 19th. It transformed the western world from a rural society to an urban one, and set the foundations for modern capitalist society.

The term "Industrial Revolution" implies a sudden and universal change, but it was not a single event, rather a series of technological innovations, social developments, and economic growth spurts that fed into each other. Britain was fortunate in having the conditions it needed to be the industrial pioneer. Firstly, there was a natural abundance of the raw materials: water, iron, and coal. Secondly, there was available capital: interest rates were low, and members of an increasingly wealthy middle

class, looking for ways to invest their money, were eager to support new inventions and technology. Finally, Britain had an unusually large market for its manufactured goods, with an expanding empire, and a dominance

of the seas, along with a strong merchant navy to trade and transport goods around the world.

The steam engine powered the revolution. Invented by Thomas Newcomen in 1712, the improvements

"I sell here, Sir, what the world desires to have—power."

MATTHEW BOULTON, BRITISH ENGINEER, 1776